STRUCTURAL DYNAMICS

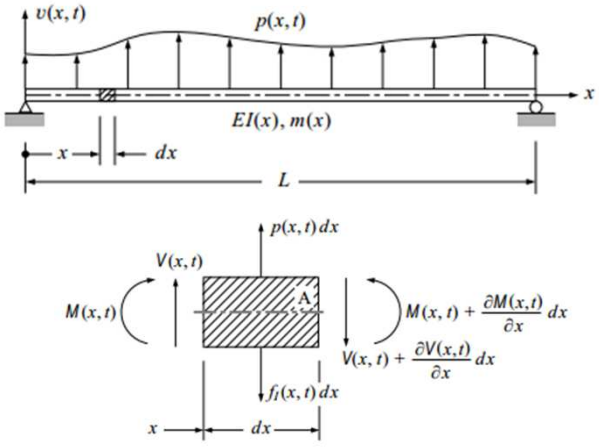
4.0 Continuous System (4 hours)

4.1 Partial differential equations of motion (Transverse vibration of a string and beam; axial vibration of a bar)
4.2 Evaluation of natural frequencies and mode shapes (Transverse vibration of a string and beam)

4.1 Partial differential equations of motion (for String, Beam, Bar)

Transverse vibration of a beam:

Considering equilibrium of forces acting on the differential segment of a beam in vertical direction.



$$V(x,t) + p(x,t)\,dx - \left[V(x,t) + \frac{\partial V(x,t)}{\partial x}\,dx\right] - f_I(x,t)\,dx = 0$$

4.1 Partial differential equations of motion (for String, Bar, Beam)

Where, $$f_I(x,t)\,dx = m(x)\,dx\,\frac{\partial^2 v(x,t)}{\partial t^2}$$

Transverse inertial force, (mass x acceleration).

4.1 Partial differential equations of motion (for String, Bar, Beam)

$\therefore p(x,t)\,dx - \frac{\partial V(x,t)}{\partial x}dx - m(x)dx\,\frac{\partial^2 v(x,t)}{\partial t^2} = 0$

$\Rightarrow \frac{\partial V(x,t)}{\partial x} = p(x,t) - m(x)\frac{\partial^2 v(x,t)}{\partial t^2}$ ——— (1)

Summing moments about point 'A'.

$M(x,t) + V(x,t)\cdot dx + p(x,t)\cdot dx\cdot \frac{dx}{2} - f_I(x,t)\,dx\cdot\frac{dx}{2} - \left[M(x,t) + \frac{\partial M(x,t)}{\partial x}dx\right] = 0$

$\Rightarrow \frac{\partial M(x,t)}{\partial x} = V(x,t)$ ——— (2)

Differentiating this eqn w.r.t. x

$\frac{\partial V(x,t)}{\partial x} = \frac{\partial^2 M(x,t)}{\partial x^2}$ ——— (3)

4.1 Partial differential equations of motion (for String, Bar, Beam)

From eqns ① & ③ ⑧

$$\frac{\partial^2 M(x,t)}{\partial x^2} + m(x)\frac{\partial^2 v(x,t)}{\partial t^2} = p(x,t) \quad \text{—④}$$

Introducing basic moment-curvature relationship

$M = EI\frac{\partial^2 v}{\partial x^2}$, becomes,

$$\frac{\partial^2}{\partial x^2}\left[EI(x)\frac{\partial^2 v(x,t)}{\partial x^2}\right] + m(x)\frac{\partial^2 v(x,t)}{\partial t^2} = p(x,t) \quad \text{—⑤}$$

This is partial differential equation of motion for the elementary case of beam flexure.

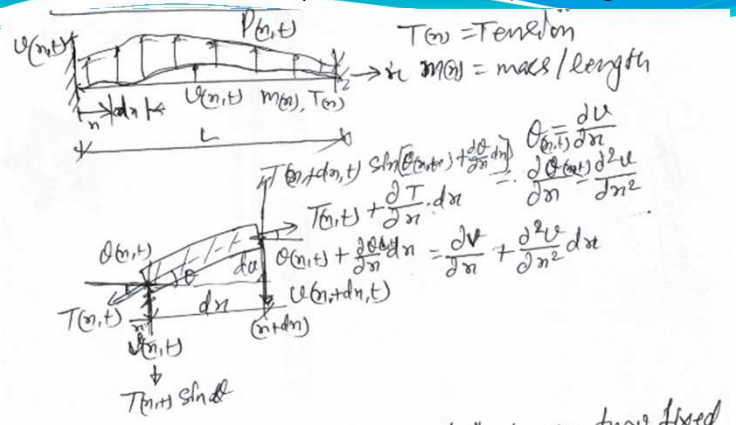
Transverse vibration of string:

Boundary conditions:

$v(0,t) = v(L,t) = 0$,

$\frac{\partial^2 v}{\partial x^2} = 0$,

4.1 Partial differential equations of motion (for String, Bar, Beam)



Consider a string stretched between two fixed points at $x = 0$ and at $x = L$.

Assume, S = x-sectional area of string.

ΔL = initial stretch

Then initial tension, $T = ES\frac{\Delta L}{L}$

4.1 Partial differential equations of motion (for String, Bar, Beam)

~~Moments are 0~~ ✓ ✓ for small angle, $\theta \simeq \sin\theta$

Equating the all vertical forces to zero we get,

$$-T(x,t)\cdot\sin\theta_{(x,t)} + \left\{T(x,t) + \frac{\partial T}{\partial x}dx\right\}\sin\left[\theta_{(x,t)} + \frac{\partial \theta_{(x,t)}}{\partial x}dx\right] + P(x,t)\,dx - f_{I(x,t)}\,dx = 0$$

But $f_{I(x,t)} = m(x)\,dx\cdot\frac{\partial^2 v}{\partial t^2}$

$$\therefore -\cancel{T(x,t)\cdot\frac{\partial v}{\partial x} + T(x,t)\cdot\frac{\partial v}{\partial x}} + \cdots$$

$$-T(x,t)\cdot\frac{\partial v}{\partial x} + \left[T(x,t) + \frac{\partial T}{\partial x}dx\right]\times\left[\frac{\partial v}{\partial x} + \frac{\partial^2 v}{\partial x^2}dx\right] + P(x,t)\,dx$$

$$T\cdot\frac{\partial^2 v}{\partial x^2}dx + \cdots - m(x)\,dx\cdot\frac{\partial^2 v}{\partial t^2} = 0$$

4.1 Partial differential equations of motion (for String, Bar, Beam)

$$T(x,t)\cdot\frac{\partial^2 v}{\partial x^2}dx + \cdots - m(x)\,dx\cdot\frac{\partial^2 v}{\partial t^2} = 0$$

$$\text{i.e. } \frac{\partial T}{\partial x}dx\cdot\frac{\partial v}{\partial x} + \frac{\partial T}{\partial x}dx\cdot\frac{\partial^2 v}{\partial x^2}dx + P(x,t)\,dx - m(x)\,dx\cdot\frac{\partial^2 v}{\partial t^2} = 0$$

$$\text{i.e. } m(x)\frac{\partial^2 v}{\partial t^2} - T(x,t)\cdot\frac{\partial^2 v}{\partial x^2} - \frac{\partial T}{\partial x}\cdot\frac{\partial v}{\partial x} - \frac{\partial T}{\partial x}dx\cdot\frac{\partial^2 v}{\partial x^2} - P(x,t) = 0$$

For small vibration, it is reasonable to assume that $T(x,t) = T(x)$ = initial tension.
(dynamic tension is neglected)

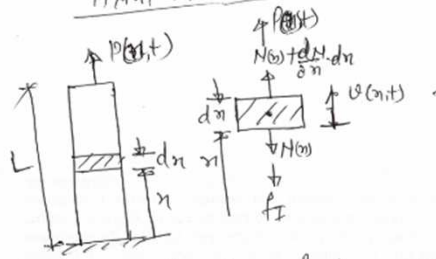
∴ And for an uniform string.

$T(x) = T$, $m(x) = m$

$$\therefore m\frac{\partial^2 v}{\partial t^2} - T\cdot\frac{\partial^2 v}{\partial x^2} - P(x,t) = 0 \quad \text{——(6)}$$

Boundary conditions, $v(0,t) = v(L,t) = 0$

Axial vibration of a bar.



N → axial force

$f_I(x,t) = m \cdot dx \cdot \frac{\partial^2 v}{\partial t^2}$

from equilibrium of the segment,

$$\therefore N + \frac{\partial N}{\partial x} dx - N - f_I + p(x,t) = 0$$

$$\frac{\partial N}{\partial x} dx - m\,dx \cdot \frac{\partial^2 v}{\partial t^2} + p(x,t) = 0$$

For free vibration $p(x,t) = 0$

$$\boxed{m \frac{\partial^2 v}{\partial t^2} - \frac{\partial N}{\partial x} = 0} \quad (7)$$

ϵ - strain = change in length / original length

$$\boxed{m \frac{\partial^2 v}{\partial t^2} - \frac{\partial N}{\partial x} = 0}$$

From mechanics of bar,

ϵ = strain = change in length / original length

$$N = \sigma \cdot A = E\epsilon \cdot A = EA \cdot \frac{\partial v}{\partial x} \qquad = \frac{\partial v}{\partial x}$$

$$\therefore N = EA \frac{\partial v}{\partial x} \quad (8)$$

Substituting in eqn (7) we get,

$$m \frac{\partial^2 v}{\partial t^2} - EA \frac{\partial^2 v}{\partial x^2} = 0 \quad (9)$$

Boundary conditions:

Fixed end: $v(0,t) = 0$

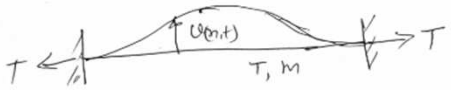
Free end: $\frac{\partial v(L,t)}{\partial x} = 0$, when $P(t) = 0$

$\frac{\partial v(L,t)}{\partial x} = \frac{P(t)}{EA}$, when $P(t) \neq 0$

4.2 Evaluation of natural frequencies and mode shapes (Transverse Vibration of String and Beam)

Transverse vibration of a string:- (11)

Natural frequencies and mode shapes:-



Vibration dynamic equation of string,

$$m\frac{\partial^2 v}{\partial t^2} - T\frac{\partial^2 v}{\partial x^2} - p(x,t) = 0 \quad \text{—— (6)}$$

For free vibration,

$$m\frac{\partial^2 v}{\partial t^2} - T\frac{\partial^2 v}{\partial x^2} = 0 \quad \text{—— (10)}$$

Displacement can be represented by mode shape and amplitude. i.e.

$$v(x,t) = \phi(x) \cdot Y(t) \quad \text{—— (11)}$$

$\phi(x)$ → mode shape

$Y(t)$ → amplitude

4.2 Evaluation of natural frequencies and mode shapes (Transverse Vibration of String and Beam)

$$\therefore \frac{\partial v(x,t)}{\partial x} = \phi'(x)\,Y(t) \quad \text{and} \quad \frac{\partial v(x,t)}{\partial t} = \phi(x)\,\dot{Y}(t)$$

$$\text{and,} \quad \frac{\partial^2 v(x,t)}{\partial x^2} = \phi''(x)\,Y(t) \quad \text{and} \quad \frac{\partial^2 v(x,t)}{\partial t^2} = \phi(x)\,\ddot{Y}(t) \quad \text{—— (12)}$$

Substituting in eqn (10)

$$m \cdot \phi(x)\,\ddot{Y}(t) - T \cdot \phi''(x) \cdot Y(t) = 0$$

Dividing by $\phi(x) \cdot Y(t)$ we get,

$$m\frac{\ddot{Y}(t)}{Y(t)} - T\frac{\phi''(x)}{\phi(x)} = 0$$

$$\text{or,} \quad \frac{\ddot{Y}(t)}{Y(t)} = \left(\frac{T}{m}\right)\cdot\frac{\phi''(x)}{\phi(x)} \quad \text{—— (13)}$$

Here,

T/m – is a constant

$\ddot{Y}(t)/Y(t)$ → is function of time

$\phi''(x)/\phi(x)$ → is function of ~~x~~ or (~~x~~ position)

4.2 Evaluation of natural frequencies and mode shapes (Transverse Vibration of String and Beam)

Assume, $\frac{T}{m} = c^2$ (12)

Because first term in this equation is a function of x only and the second term is a function of t only, the entire equation can be satisfied for arbitrary values of x and t only, if each term is a constant in accordance with,

$$\frac{\ddot{Y}(t)}{Y(t)} = c^2 \frac{\phi''(x)}{\phi(x)} = -\omega^2$$

$$\therefore \ddot{Y}(t) + \omega^2 Y(t) = 0 \quad \& \quad \phi''(x) + \frac{\omega^2}{c^2}\phi(x) = 0 \quad \text{— (14)}$$

Both are ordinary differential equations.

Let, $\frac{\omega}{c} = \lambda \therefore \phi''(x) + \lambda^2 \phi(x) = 0$

Solution of eqn (14) is given by,

4.2 Evaluation of natural frequencies and mode shapes (Transverse Vibration of String and Beam)

$$Y(t) = A\cos\omega t + B\sin\omega t \quad \text{— (15)}$$

and $\phi(x) = C\sin\lambda x + D\cos\lambda x \quad \text{— (16)}$

Considering the boundary conditions

$u(0,t) = \phi(0)\,Y(t) = 0$, implies, $\phi(0) = 0$

$u(L,t) = \phi(L)\,Y(t) = 0$, implies, $\phi(L) = 0$

From eqn (16),

$$\phi(0) = C\sin\lambda\cdot 0 + D\cos\lambda\cdot 0 = 0$$

$$\therefore D = 0$$

Also, $\phi(L) = C\sin\lambda L + D\cos\lambda L = 0$

$C\cdot\sin\lambda L = 0 \Rightarrow C \neq 0$

Expressing in matrix form,

$$\begin{bmatrix} 0 & 1 \\ \sin\lambda L & \cos\lambda L \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad \text{— (17)}$$

4.2 Evaluation of natural frequencies and mode shapes (Transverse Vibration of String and Beam)

D At least one of the variable of C & D should (137)(13) not be equal to zero.

Since $\begin{bmatrix} C \\ D \end{bmatrix}$ is not equal to zero, determinant of $\begin{bmatrix} 0 & 1 \\ \sin\lambda L & \cos\lambda L \end{bmatrix}$ should be zero for non-trivial solution.

i.e. $\begin{vmatrix} 0 & 1 \\ \sin\lambda L & \cos\lambda L \end{vmatrix} = 0$

$\therefore \sin\lambda L = 0$ — (18)

$\therefore \lambda L = \pi, 2\pi, 3\pi, \dots\dots n\pi$

$\therefore \boxed{\lambda_n = \frac{n\pi}{L}}$ — (19) for $n = 1, 2, \dots\dots n$

4.2 Evaluation of natural frequencies and mode shapes (Transverse Vibration of String and Beam)

we have, $\lambda_n = \frac{\omega_n}{c} = \frac{n\pi}{L}$

$\therefore \boxed{\omega_n = \frac{n\pi}{L} \times \sqrt{\frac{T}{m}}}$ — (20)

from eqn (16)

$\phi_n(x) = C \sin\left(\frac{n\pi}{L} x\right)$ for $n = 1, 2, 3, \dots\dots n$

Here, C can be any variable. So let $C = 1$

$\therefore \phi_{n(x)} = \sin\frac{n\pi x}{L}$ — (21) for, $n = 1, 2, 3, \dots$

When $n = 1$

$\phi_{1(x)} = \sin\frac{\pi x}{L}$



$\omega_1 = \frac{\pi}{L}\sqrt{\frac{T}{m}}$

4.2 Evaluation of natural frequencies and mode shapes (Transverse Vibration of String and Beam)

When, $n=2$ (14)

$\phi_2(x) = \sin\frac{2\pi x}{L}$  $\omega_2 = \frac{2\pi}{L}\sqrt{\frac{T}{m}}$

When, $n=3$

$\phi_3(x) = \sin\frac{3\pi x}{L}$  $\omega_3 = \frac{3\pi}{L}\sqrt{\frac{T}{m}}$

From eqn (11)

$$y(x,t) = \sum_{i=1}^{\infty}\left(A_i \cos\omega_i t + B_i \sin\omega_i t\right)\left(\sin\frac{i\pi x}{L}\right) \quad \text{——(22)}$$

At initial state ($t=0$)

$$y(x,0) = \sum_{i=1}^{\infty} A_i \sin\frac{i\pi x}{L} \quad \text{——(23)}$$

4.2 Evaluation of natural frequencies and mode shapes (Transverse Vibration of String and Beam)

Multiplying eqn (23) by $\sin\frac{i\pi x}{L}$ and integrate over the length of the string, we get.

$$\int_0^L y(x,0)\cdot\sin\frac{i\pi x}{L}\,dx = \int_0^L\left(\sum_{i=1}^{\infty} A_i \sin\frac{i\pi x}{L}\right)\cdot\sin\frac{i\pi x}{L}\,dx$$

$$\text{or, } \int_0^L y(x,0)\cdot\sin\frac{i\pi x}{L}\,dx = \frac{A_i}{2}L$$

$$\therefore A_i = \frac{2}{L}\int_0^L y(x,0)\cdot\sin\left(\frac{i\pi x}{L}\right)dx \quad \text{——(24)}$$

initial configuration.

In similar manner,

$$\dot{y}(x,t) = \sum_{i=1}^{\infty}\left(-A_i\omega_i \sin\omega_i t + B_i\omega_i\cos\omega_i t\right)\cdot\sin\frac{i\pi x}{L}$$

at $t=0$, $\dot{y}(x,t) = \sum_{i=1}^{\infty}\omega_i B_i \cdot \sin\frac{i\pi x}{L}$

4.2 Evaluation of natural frequencies and mode shapes (Transverse Vibration of String and Beam)

$$\int_0^L \dot{v}(x,t)\cdot \sin\frac{i\pi x}{L}dx = \int_0^L \sum_{i=1}^{\infty} \omega_i B_i \cdot \sin^2\left(\frac{i\pi x}{L}\right) dx$$

$$\text{or } \int_0^L \dot{v}(x,t)\cdot \sin\frac{i\pi x}{L}dx = \frac{\omega_i B_i L}{2}$$

$$\therefore B_i = \frac{2}{\omega_i L}\int_0^L \dot{v}(x,t)\cdot \sin\frac{i\pi x}{L}dx \quad \text{—— (25)}$$

Natural frequencies and mode shapes of Beam :-

Free vibration eqn of beam,

$$m(x)\frac{\partial^2 v}{\partial t^2} + \frac{\partial}{\partial x^2}\left(EI_x \frac{\partial^2 v}{\partial x^2}\right) = 0 \quad \text{—— (26)}$$

Attempt a solution of the form

4.2 Evaluation of natural frequencies and mode shapes (Transverse Vibration of String and Beam)

$$v(x,t) = \phi(x)\cdot Y(t) \quad \text{—— (27)}$$

then, $\frac{\partial^2 v}{\partial t^2} = \phi(x)\ddot{Y}(t)$ & $\frac{\partial^2 v}{\partial x^2} = \phi''(x)Y(t)$ —— (28)

$$\therefore -\frac{\ddot{Y}(t)}{Y(t)} = +\frac{EI(x)}{m(x)}\cdot\frac{\phi''(x)}{\phi(x)} \quad \text{—— (29)}$$

The two expressions must be constant, say ω^2.

$$\therefore \ddot{Y}(t) + \omega^2 Y(t) = 0$$

& $$EI(x)\cdot\phi''(x) - \omega^2 m(x)\phi(x) = 0$$

For uniform beam, $EI(x) = EI$ & $m(x) = m$

4.2 Evaluation of natural frequencies and mode shapes (Transverse Vibration of String and Beam)

$$\therefore EI\phi''(x) - \omega^2 m\phi(x) = 0 \quad (16)$$

$$\text{or } \phi''(x) - \frac{\omega^2 m}{EI}\phi(x) = 0$$

$$\text{Say } \beta^4 = \frac{\omega^2 m}{EI}$$

$$\phi''(x) - \beta^4\phi(x) = 0$$

The general solution of above equation is,

$$\phi(x) = C_1 \sin\beta x + C_2 \cos\beta x + C_3 \sinh\beta x + C_4 \cosh\beta x \quad (30)$$

For uniform, simply supported beam, at $x = 0$ and at $x = L$, the displacement and bending moment are zero.

$$\therefore v(0,t) = 0 \Rightarrow \phi(0) = 0 \Rightarrow C_2 + C_4 = 0$$

$$M(0,t) = 0 \Rightarrow EI\frac{\partial^2 v}{\partial x^2} = EI\phi''(0) = 0 \Rightarrow \beta^2(-C_2 + C_4) = 0$$

4.2 Evaluation of natural frequencies and mode shapes (Transverse Vibration of String and Beam)

$\therefore C_2 = C_4 = 0$ and general solution reduces to,

$$\phi(x) = C_1 \sin\beta x + C_3 \sinh\beta x \quad (31)$$

At $x = L$,

$$v(L,t) = 0 \Rightarrow \phi(L) = 0 \Rightarrow C_1 \sin\beta L + C_3 \sinh\beta L = 0$$

& $$M(L,t) = 0 \Rightarrow EI\phi''(L) = 0 \Rightarrow \beta^2(-C_1 \sin\beta L + C_3 \sinh\beta L) = 0$$

adding these two eqn.s,

$$C_3 \sinh\beta L = 0$$

Since $\sinh\beta L$ can not be zero (otherwise 'w' will be zero, a trivial solution implying no vibration at all), so C_3 must be zero. This leads to the frequency equation $C_1 \sin\beta L = 0$

4.2 Evaluation of natural frequencies and mode shapes (Transverse Vibration of String and Beam)

This equation can be satisfied by selecting (IT)
$C_1 = 0$, which gives $\phi(x) = 0$, a trivial solution.
Therefore, $\sin \beta L$ must be zero and,

$$\beta L = n\pi, \quad n = 1, 2, 3, \text{----}$$

or, $$\beta = \frac{n\pi}{L}$$

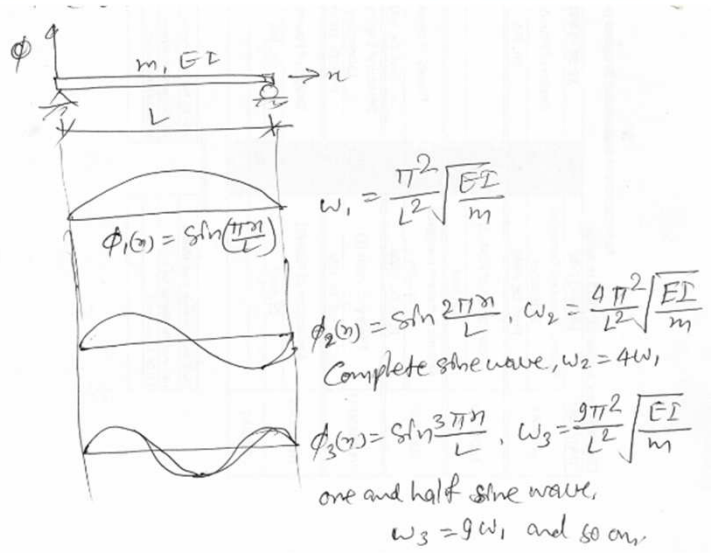
and, $$\frac{\omega_n^2 m}{EI} = \beta^4 = \left(\frac{n\pi}{L}\right)^4$$

or $$\omega_n = \frac{n^2\pi^2}{L^2}\sqrt{\frac{EI}{m}}, \quad n = 1, 2, 3, \text{----}$$

$$\therefore \phi_n(x) = C_1 \sin\frac{n\pi x}{L} \quad \text{---- (32)}$$

The value of C_1 is arbitrary; Assume $C_1 = 1$

4.2 Evaluation of natural frequencies and mode shapes (Transverse Vibration of String and Beam)



$$\omega_1 = \frac{\pi^2}{L^2}\sqrt{\frac{EI}{m}}$$

$$\phi_2(x) = \sin\frac{2\pi x}{L}, \quad \omega_2 = \frac{4\pi^2}{L^2}\sqrt{\frac{EI}{m}}$$

Complete sine wave, $\omega_2 = 4\omega_1$

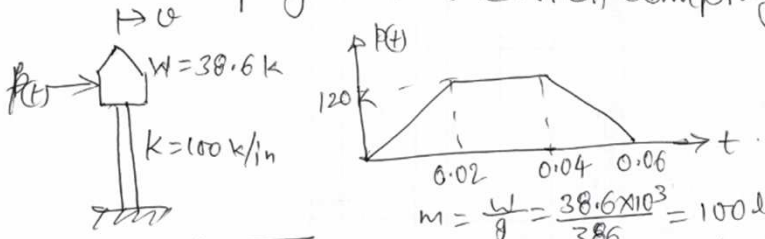
$$\phi_3(x) = \sin\frac{3\pi x}{L}, \quad \omega_3 = \frac{9\pi^2}{L^2}\sqrt{\frac{EI}{m}}$$

one and half sine wave,
$\omega_3 = 9\omega_1$ and so on.

4.2 Evaluation of natural frequencies and mode shapes (Transverse Vibration of String and Beam)

(18)

Ex.
Determine the dynamic response of a tower subjected to a blast loading. The idealization of structure and blast loading are shown. Assume damping = 20% of critical damping.



$$m = \frac{W}{g} = \frac{38.6 \times 10^3}{386} = 100 \text{ lb-s}^2/\text{in}$$

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{100{,}000}{100}} = 31.623 \text{ rad/sec} \qquad \xi = 20\% = 0.02$$

$$\omega_D = \omega\sqrt{1-\xi^2} = 31.623\sqrt{1-(0.2)^2} = 30.984 \text{ rad/sec}$$

4.2 Evaluation of natural frequencies and mode shapes (Transverse Vibration of String and Beam)

$$\therefore T = \frac{2\pi}{\omega} = \frac{2\pi}{31.623} = 0.20 \text{ sec.}$$

Select $\Delta t \le \frac{T}{10} = \frac{0.20}{10} = 0.02$ sec.

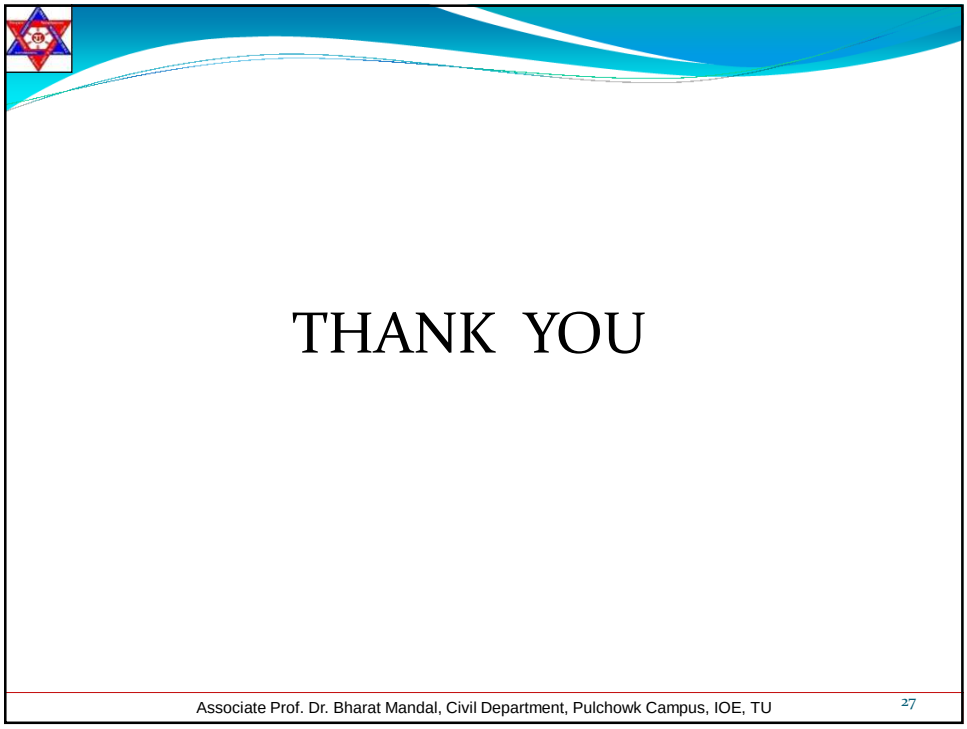
$$\int_0^{t_1} P(t)\,dt = \tfrac{1}{2} \times 120 \times 0.02 + 120 \times (0.04 - 0.02) + \tfrac{1}{2} \times 120 \times 0.02$$
$$= 1.2 + 2.4 + 1.2 = 4.8 \text{ k}$$

Response then approximately,

$$v(t) = \frac{p_0}{m\omega^2}\sin\omega t = \frac{p_0}{m\omega}\sin\omega t$$
$$= \frac{4.8 \times 1000}{100 \times 31.623}\sin\omega t = 1.5179 \sin\omega t$$

$$\therefore v_{max} = 1.5179 \text{ in}$$

max elastic force, $f_{s\,max} = k \times v_{max}$.


THANK YOU
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